Unsupervised Hybrid Network of Transformer and CNN for Blind Hyperspectral and Multispectral Image Fusion

Xuheng Cao, *Student Member, IEEE*, Yusheng Lian, Kaixuan Wang, Chao Ma, and Xianqing Xu

***Abstract*— Fusing a low spatial resolution hyperspectral image (LR-HIS) with a high spatial resolution multispectral image has become popular for generating a high spatial resolution hyperspectral image (HR-HSI). Most methods assume that the degradation information from high resolution to low resolution is known in spatial and spectral domains. Conversely, this information is often limited or unavailable in practice, restricting their performance. Furthermore, existing fusion methods still face the problem of insufficient exploration of the cross-interaction between the spatial and spectral domains in the HR-HSI, leaving scope for further improvement. This article proposes an unsupervised hybrid network of transformer and CNN (uHNTC) for blind HSI-MSI fusion. The uHNTC comprises three subnetworks: a transformer-based feature fusion subnetwork (FeafusFomer) and two CNN-based degradation subnetworks (SpaDNet and SpeDNet). Considering the strong multilevel spatio-spectral correlation between the desired HR-HSI and the observed images, we design a multilevel cross-feature attention (MCA) mechanism in FeafusFormer. By incorporating the hierarchical spatio-spectral feature fusion into the attention mechanism in the transformer, the MCA globally keeps a high spatio-spectral cross-similarity between the recovered HR-HSI and observed images, thereby ensuring the high cross-interaction of the recovered HR-HSI. Subsequently, the characteristics of degradation information are utilized to guide the design of the SpaDNet and SpeDNet, which helps FeafusFormer accurately recover the desired HR-HSI in complex real-world environments. Through an unsupervised joint training of the three subnetworks, uHNTC recovers the desired HR-HSI without preknown degradation information. Experimental results on three public datasets and a WorldView-2 images show that the uHNTC outperforms ten state-of-the-art fusion methods. Code available: https://github.com/Caoxuheng/HIFtool.**

***Index Terms*— Blind fusion, degradation representation, feature fusion, superresolution, unsupervised transformer.**

I. Introduction

MULTISENSOR image fusion has been widely used in image acquisition and enhancement [1], [2]. Due to the hardware limitations of existing imaging sensors, there is a tradeoff between spatial resolution and spectral resolution, resulting in a low spatial resolution of hyperspectral imaging devices. Superresolution technology [3], [4], [5], [6], especially the fusion of a low spatial resolution hyperspectral image (LR-HSI) and a high spatial resolution multispectral image (HR-MSI), has been widely used for the HR-HSI imaging due to its improved performance in many fields, such as anomaly detection [7], and image classification [8].

Generally, according to the physical imaging mechanism, the LR-HSI and HR-MSI are considered degradation versions of the HR-HSI in the spatial and spectral domains, respectively. By predefining the spatial and spectral degradation information, early works [9], [10], [11] formulated the HSI-MSI fusion as an optimization problem constrained by different hand-crafted priors. In practice, the predefined degradation models do not always hold the truth, and the imprecise degradation might lead to failure to recover the desired HR-HSI. More importantly, the hand-crafted priors always show an insufficient representation of desired HR-HSI, which limits their reconstruction accuracy.

Deep learning provides a breakthrough in HSI-MSI fusion due to its efficient feature exploration ability. Some supervised methods based on CNN directly learned the nonlinear mapping from observed images to the desired HR-HSI without degradation models [12], [13], [14]. Without guidance from the physical imaging model, their deep networks are always black boxes or lack transparency, leaving room for further improvement. Consequently, by incorporating the physical imaging model into the deep network, deep unfolding-based methods [15], [16], [17] achieved great attention as impressive performance. These methods are trained in a supervised manner, which can learn the degradation models from numerous triple data pairs (i.e., the HR-HSI, LR-HSI, and the HR-MSI). Owing to the severe data scarcity, the extensive training samples and complex network architectures make them lose practicality [18]. Furthermore, once the network is trained successfully, the degradation information is stationary, reducing their generalization [19].

Recently, unsupervised learning that reconstructs the HR-HSI from corresponding HSI-MSI pairs provides a feasible solution for data scarcity. Existing unsupervised methods [20], [21], [22] relied on predefined spatial and spectral degradation models. When coping with the unknown degradation, these methods show significant performance drop or even failure. Therefore, it is essential to investigate an

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The authors are with the School of Printing and Packaging Engineering, Beijing Institute of Graphic Communication, Beijing 102600, China (e-mail: caoxuheng@tongji.edu.cn; lianyusheng@126.com; 15510201316@163.com; mc2953470349@163.com; xuxianqingbg@163.com).

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unsupervised blind method without using predefined degradation models.

Dealing with the unknown degradation is a challenge for unsupervised method. The naive solution is to directly utilize trainable parameters to simulate the degradation models. By employing a convolutional or linear layer, works [15], [23], [24], [25], [26], [27], [28] attempted to learn the spatial and spectral degradation models from the observed images. Since the high complexity of the real-world degradation, such a simple simulation is always unsuitable [18]. Also, the characteristics of the degradation information (e.g., the nonnegativity and unit dc gain) is ignored, which causes their network training to falls into a local optimum [29]. Consequently, Xu et al. [19] employed a Resblock to learn the spatial degradation, and utilized the SoftMax layer to guarantee the nonnegative and normalization of the degradation model. However, the degradation presentation in the structure region is the key to spatial degradation, which the Resblock cannot efficiently model. In addition, the smooth property of the spectral degradation is ignored. Therefore, it is meaningful to design the degradation networks with the guidance of the characteristics of degradation information.

Another challenge for unsupervised blind methods lies in the insufficient feature exploration of the HR-HSI. Specifically, existing unsupervised methods [15], [23], [24], [25], [26], [27], [28] were designed based on CNN, which entirely relies upon the convolutional layer to explore the local features. Although work [19] introduced the SwinBlock as a fusion block, the swin-transformer was built for discarding irrelevant information to exact high-level visual features, thus making it hard to preserve the inherent property in HSI, e.g., the interspectra similarities. Conversely, supervised transformers [30], [31] have realized the importance of feature exploration, and attempted to solve this problem by applying self-attention to spectral and spatial domains. These supervised methods with succeeding transformer blocks lack the appropriate guidance over the intermediate features within the network, causing an inadequate exploitation of observed images. Furthermore, their attention mechanisms independently explore the HR-HSI feature in the spatial and spectral domains, which cannot efficiently explore the cross-interaction between two domains, ultimately causing an inevitable spatio-spectral distortion [32].

To address the above problems, we propose an unsupervised hybrid network of transformer and CNN (uHNTC). As shown in Fig. 1, the proposed uHNTC mainly comprises two CNN-based degradation subnetworks and a transformer-based fusion subnetwork. The spatial degradation network (SpaDNet) and the spectral degradation network (SpeDNet) are designed following the characteristics of degradation information. In SpaDNet, a dual path architecture is proposed to represent spatial degradation, in which one path represents the structural degradation, and another path represents the smooth degradation. The dual path design enables the SpaDNet to pay well attention to the degradation of structural regions, which can improve its degradation representation ability in coping with complex real-world environments. Subsequently, SpeDNet employs a 1-D convolutional layer to adjust the weights in a linear layer, which can efficiently exploit the latent

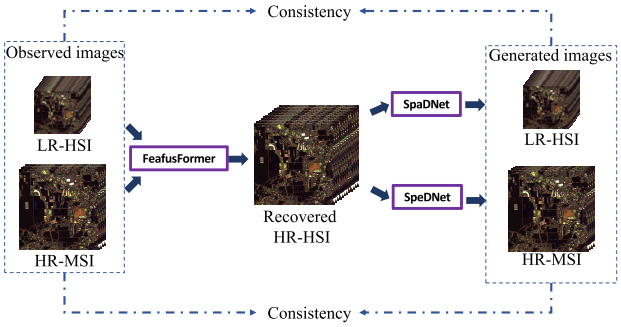


Fig. 1. Illustration of the proposed unsupervised blind fusion method.

smooth characteristics of the spectral degradation. Also, to further exploit the sparsity and smoothness of the degradation information, we investigate a subspace-based loss function, significantly improving the representation accuracy of degradation networks.

In the fusion subnetwork FeafusFormer, a novel unsupervised transformer-based fusion idea is proposed. Generally, the desired HR-HSI is highly similar to the LR-HSI and HR-MSI in the spatial and spectral domains. Consequently, the motivation behind FeafusFormer is to model the cross-similarity in the spatio-spectral domain between the desired HR-HSI and the observed images, naturally ensuring the spatio-spectral cross-interaction in the desired HR-HSI. Furthermore, considering that the multilevel features of the LR-HSI and HR-MSI retain rich spectral and spatial features, we propose that the hierarchical features of observed images are employed to guide the HR-HSI recovery in a level-by-level manner gradually. This framework, with the guidance of multilevel features, makes full use of the observes' hierarchical feature in each level and takes appropriate guidance over the intermediate features within the network, ensuring high spatio-spectral feature similarity between the recovered HR-HSI and the observes. Accordingly, we design a multilevel cross-feature attention (MCA) to recover the desired hierarchical feature by efficiently exploring the interactions between the hierarchical features of LR-HSI and HR-MSI, thereby ensuring the cross-interaction in the recovered HR-HSI.

Finally, leveraging the consistency between the observed images and generated images (see Fig. 1), uHNTC recovers the desired HR-HSI with blind degradations in an unsupervised closed-form manner.

The main contributions are summarized as follows.

1) A novel unsupervised hybrid network is proposed for blind HSI-MSI fusion. The proposed network can recover the HR-HSI without predefining degradation models.
2) Following the characteristics of the degradation information, the spatial and spectral degradation networks are designed. In addition, a subspace-based loss function is proposed to further improve the representation accuracy of degradation subnetworks.
3) A novel multilevel feature fusion transformer, FeafusFormer, is proposed. In the FeafusFormer, the HR-HSI recovery is guided by the hierarchical

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features of observes, which enables it to fully exploit the spatio-spectral features of the observes, effectively reducing spatio-spectral distortion.

4) An MCA mechanism is proposed to model the global spatio-spectral cross-similarity between the desired HR-HSI and the observed images, which can keep the high similarity between the recovered HR-HSI and observed images, thereby naturally ensuring the cross-interaction of the recovered HR-HSI.

The remaining part of this article is organized as follows. Section II introduces the related works, including the observation model, nonblind methods, and blind methods. Section III describes the proposed network architecture and the loss function. Section IV shows the ablation studies and experimental results on simulation datasets and real-world applications. Finally, the conclusion is given in Section V.

II. Related Works

A. Observation Model

The HSI-MSI fusion problem requires solving the desired HR-HSI $\mathbf{Z} \in \mathbb{R}^{W\times H\times C}$, where W, H, and C are the weight, height, and the number of spectral bands, respectively. The observed images include an HR-MSI $\mathbf{Y} \in \mathbb{R}^{W\times H\times c}$ and an LR-HSI $\mathbf{X} \in \mathbb{R}^{w\times h\times C}$, where $W = wr$, $H = hr$, and r is the scale factor. In this article, $r = 32$. To simplify notation, we unfold the 3D-tensors $\mathbf{Z}$, $\mathbf{Y}$, and $\mathbf{X}$ to form 2D-matrices $\hat{\mathbf{Z}} \in \mathbb{R}^{C\times WH}$, $\hat{\mathbf{Y}} \in \mathbb{R}^{c\times WH}$, and $\hat{\mathbf{X}} \in \mathbb{R}^{C\times wh}$. Each column in $\hat{\mathbf{Z}}$ represents the spectrum vector for a pixel. In particular, the physical imaging model considers the HR-MSI $\hat{\mathbf{Y}}$ is spectrally degraded from the HR-HSI $\hat{\mathbf{Z}}$, and the LR-HSI $\hat{\mathbf{X}}$ is spatially degraded from the HR-HSI $\hat{\mathbf{Z}}$. In mathematics, their correlation can be formulated as follows:

$$\hat{\mathbf{Y}} = \hat{\mathbf{R}}\mathbf{Z}, \quad \hat{\mathbf{X}} = \mathbf{Z}\hat{\mathbf{B}}\mathbf{S} \tag{1}$$

where $\mathbf{R} \in \mathbb{R}^{3\times C}$ is the spectral response function (SRF), which denotes the spectral degradation, and $\mathbf{BS} \in \mathbb{R}^{MN\times mn}$ denotes spatial degradation. Matrix $\mathbf{B}$ is a blurring matrix employed to simulate the point spread function, and matrix $\mathbf{S}$ is a downsample matrix. The spectral and spatial degradations are predefined in the nonblind fusion task. The fusion methods without predefining degradation models belong to blind methods.

B. Nonblind Methods

Various nonblind works have been specially designed for fusion tasks, including model-based and deep learning-based methods. Most model-based methods [33], [34] assume each spectral pixel can be represented as a linear combination of spectral basis and coefficients. They recover the desired HR-HSI by reconstructing the spectral basis and coefficient matrix. For example, Dong et al. [35] solve the fusion problem with a sparse regularization, which assumes the coefficients are sparse. Zhou et al. [36] and Yokoya et al. [37] employ coupled nonnegative matrix factorization schemes to update spectral basis and coefficients alternately. Cao et al. [18], Simões et al. [34], and Pan and Shen [38] utilize the spatial edge structure of HR-MSI to guide coefficient reconstruction. These methods unfold the observed images as 2D-matrices, which cannot fully exploit the inherent HSI spatial structure. To alleviate it, tensor factorization-based methods consider the HR-HSI as a 3-D tensor. Li et al. [39] solve the fusion problem by coupled sparse tensor factorization. In LTMR, Dian and Li [40] collect the same coefficients in the same cluster into a 3-D tensor and impose low tensor multirank regularization, which can exploit the nonlocal self-similarity of the HR-HSI.

Recently, deep learning has had a breakthrough in fusion tasks. Dian et al. [41] employ a residual neural network to learn the spatio-spectral image priors. Furthermore, CNN-Fus [42] utilizes a well-trained CNN denoiser to regularize the estimation of spatial structure. Dong et al. [43] propose a model-guided deep neural network (MoG-DCN) in which the degradation prior can guide the deep CNN to recover the desired HR-HSI.

These nonblind methods assume that the spatial and spectral degradation models are known. The predefined degradation models are often unobtainable and somewhat inaccurate [22]. Therefore, these nonblind fusion processes can only be regarded as a rough approximation of the real-world observation process, which restricts their practice performance.

Various semiblind methods try to learn or skip part degradation. Qu et al. [20] recover the HR-HSI in a latent space by using an encoder–decoder-based network coupled with a sparse Dirichlet distribution, which does not need the spatial degradation model. Pan and Shen [38] estimate the spectral degradation model from the LR-HSI and HR-MSI. Dian et al. [44] employ spectra clustering to skip the spatial degradation model. Cao et al. [45] utilize HR-MSI multilevel features to guide the HR-HSI reconstruction, which does not need the spatial degradation model. HyMS [18] proposes a multistage scheme to skip spatial degradation. These semiblind methods need one of the degradation models, and their performance relies on the accuracy of the necessary degradation model.

C. Blind Methods

Recently, blind methods have attracted more attention. Model-based methods attempt to solve the spatial degradation $\mathbf{B}$ and spectral degradation $\mathbf{R}$ shown in (1). In [37], $\mathbf{B}$ is assumed to be a Gaussian blur, and its variance is estimated using a template-matching technique. Meanwhile, they estimate the $\mathbf{R}$ based on the so-called prelaunch response. HySure [34] establishes an optimization model and estimates the $\mathbf{R}$ and $\mathbf{B}$ with a regularizer. In practice, the complex degradation models could not be solved accurately, which might limit their recovery performance.

Some CNN-based methods directly learn a mapping function between the observed images $[\mathbf{Y}, \mathbf{Z}]$ and the desired $\mathbf{X}$ by employing supervised deep neural networks, such as PzResNet [13] and PanNet [46]. These methods are limited in global information exploration, resulting in heavy room for recovery accuracy improvement.

Transformer has drawn a great attention in HSI-MSI fusion as the excellent ability of its self-attention mechanism in global information exploration. FusFormer [12] first introduces the

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self-attention mechanism into HSI-MSI fusion task and obtains promising results. Due to the unbearable computational cost when applying self-attention to the whole image, FusFormer recovers the target HR-HSI patch by patch, neglecting the connection between image patches. PSRT [14] proposes a shuffle-and-reshuffle module to exploit the spatial information interaction among the image patches. FusFormer and PSRT focus on spatial information while ignoring spectral feature exploration. To capture the spatio-spectral feature, HMF-Former [30] designs a spatio-spectral Transformer block including spatial and spectral multihead attention. MSST [31] proposes a multiscale extraction architecture by self-attention applying to spatial and spectral domains with different scales. To explore the cross-interactions between spatial and spectral domains, HyperTransformer [47] improves the original global attention as a feature soft-attention that can exploit the spatio-spectral dependencies between observed images. MCT-Net [48] investigates a multihead-fusion-attention mechanism in which the token in the spatial domain is multiplied by the value in the spectral domain. Without the guidance of the physical imaging model, these supervised networks are always black boxes and lack transparency, leaving room for further improvement. Subsequently, deep unfolding provides a promising solution by incorporating the degradation characteristic into deep networks [15], [16], [17]. Although the deep unfolding-based networks can learn the degradation models from training samples, once the network is trained successfully, the degradation information is stationary. The minor changes in the imaging environment always result in spatial and spectral degradation alterations, which reduce their generalization. In addition, owing to the severe data scarcity, the extensive training samples and complex network architectures make them lose practicality.

Recently, unsupervised blind methods have drawn significant attention. These methods usually utilize a single learnable layer for degradation representation, and a shallow CNN-based network for fusing HR-MSI and LR-HSI. In DBSR, Zhang et al. [24] employ a single convolutional layer to simulate spatial degradation, a fully connected layer representing spectral degradation, and a U-Net as a fusion net to recover the desired HR-HSI. In UDALN, Li et al. [25] utilize two single-layer CNNs to represent the spatial degradation and spectral degradation, and a multilayer fully CNN is employed to fuse observed images. Zheng et al. [28] use the single convolutional layer to represent spatial and spectral degradations, and three CNN-based autoencoders are designed to recover HR-HSI. Xu et al. [19] formulate the blind problem as a maximum posterior estimation, in which the residual swin transformer recovers the HSI; the spatial and spectral degradation models are unrolled by a Resblcock and linear layer, respectively. Real-world degradation always consists of an unknown blur, noise, and compression, making it challenging to learn the degradation process using a simple network architecture. Besides, the CNN-based fusion network is so shallow that it only extracts and exploits the low-level features from observed images. Also, because the plain convolution operation cannot explicitly consider the spatial interactions between spatial locations, these CNN-based fusion networks have limited ability to exploit global information, affecting their recovery accuracy. Although work [19] employs residual swin transformers, this transformer architecture is designed for universal vision tasks, which cannot fully exploit the spatio-spectral cross-interactions in fusion tasks.

III. Methodology

We propose an uHNTC for blind HSI-MSI fusion. The uHNTC consists of three subnetworks: FeafusFomer, SpaDNet, and SpeDNet. Instead of employing known degradation models, we design SpaDNet and SpeDNet to represent the degradations. By combining CNN and Transformer, FeafusFormer fully models the cross-similarity of multilevel features between the desired HR-HSI and the observed images, and exploits the multilevel features of observed images in consideration of local and global information. In this section, we first introduce the architecture of uHNTC, SpaDNet, and SpeDNet in Section III-A. Section III-B elaborates on the architecture of the FeafusFormer.

A. Blind HSI-MSI Fusion Network uHNTC

A blind HSI-MSI fusion network uHNTC is proposed. Its overall framework is shown in Fig. 2. FeafusFormer fuses the observed images LR-HSI $\mathbf{X}$ and HR-MSI $\mathbf{Y}$ to obtain the HR-HSI $\mathbf{Z}$. By using the proposed degradation networks SpaDNet and SpeDNet, the obtained $\mathbf{Z}$ are degraded as $\mathbf{X_Z} \in \mathbb{R}^{w\times h\times C}$ and $\mathbf{Y_Z} \in \mathbb{R}^{W\times H\times 3}$, the observed images $\mathbf{Y}$ and $\mathbf{X}$ are degraded as $\mathbf{Y_Y} \in \mathbb{R}^{w\times h\times 3}$ and $\mathbf{Y_X} \in \mathbb{R}^{w\times h\times 3}$. The $\mathbf{Y_Y}$ and $\mathbf{Y_X}$ should be consistent with each other. Meanwhile, the $\mathbf{X_Z}$ and $\mathbf{Y_Z}$ should be consistent with the observed $\mathbf{X}$ and $\mathbf{Y}$, respectively. The FeafusFormer, SpaDNet, and SpeDNet are jointly trained by exploiting the abovementioned consistencies. Eventually, the desired HR-HSI is obtained by using the trained FeafusFormer. The details are displayed as follows.

First, we employ the proposed FeafusFormer to fuse the observed image HR-MSI $\mathbf{Y}$ and LR-HSI $\mathbf{X}$. Specifically, $\mathbf{X}$ and $\mathbf{Y}$ are fed into Feafusformer $Ff(\cdot)$ to obtain HR-HSI $\mathbf{Z}$:

$$Z = Ff(X, Y, \theta_{Ff}) \tag{2}$$

where θ_{Ff} is the weights in FeafusFormer.

Second, $\mathbf{Z}$ is fed into SpaDNet and SpeDNet to obtain $\mathbf{X_Z} \in \mathbb{R}^{w\times h\times C}$ and $\mathbf{Y_Z} \in \mathbb{R}^{W\times H\times 3}$:

$$\mathbf{X_Z} = \text{SpaDNet}(Z, \theta_a), \quad \mathbf{Y_Z} = \text{SpeDNet}(Z, \theta_e) \tag{3}$$

where θ_a denotes the weights in SpaDNet, θ_e denotes the weights in SpeDNet.

Meanwhile, $\mathbf{X}$ is fed into the SpeDNet to obtain $\mathbf{Y_X}$, and $\mathbf{Y}$ is fed into the SpaDNet to obtain and $\mathbf{Y_Y}$:

$$\mathbf{Y_Y} = \text{SpaDNet}(Y, \theta_a), \quad \mathbf{Y_X} = \text{SpeDNet}(\mathbf{X}, \theta_e). \tag{4}$$

Finally, the spectrally degraded version of HR-HSI $\mathbf{Y_Z}$ should be consistent with the observed $\mathbf{Y}$, the spatially degraded version of HR-HSI $\mathbf{X_Z}$ should be consistent with observed $\mathbf{X}$, and the spatially degraded version of LR-HSI $\mathbf{Y_X}$ should be consistent with the spectrally degraded version of the HR-MSI $\mathbf{Y_Y}$. Exploiting the above three consistencies, the optimal weights θ_{Ff}, θ_a, and θ_e can be obtained by

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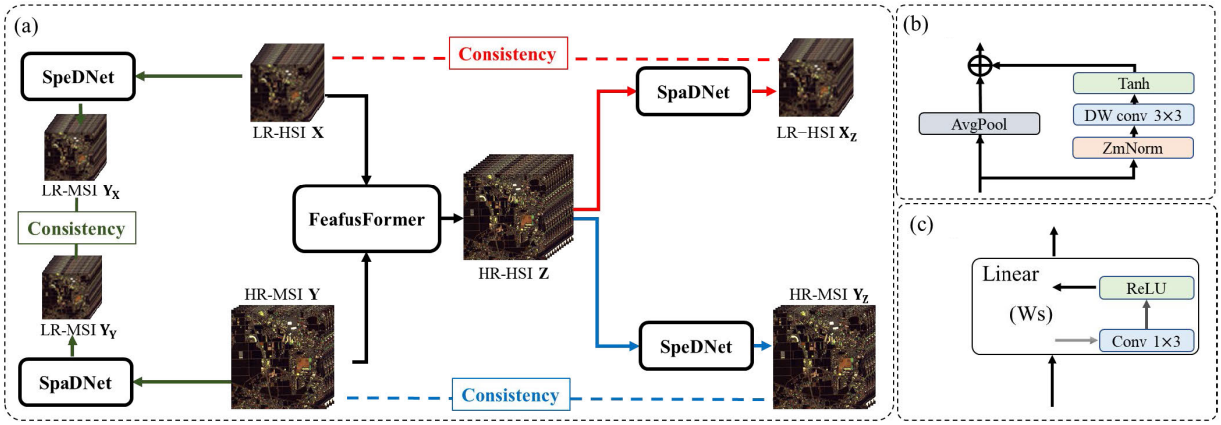


Fig. 2. (a) Framework of proposed uHNTC. (b) SpaDNet is a dual-branch CNN that can better represent more complex degradation. (c) SpeDNet can learn the smoothness prior of spectral degradation.

joint training. The desired HR-HSI $\mathbf{Z}$ is eventually obtained as (2). The training details are introduced in Section V.

The architectures of the SpaDNet and SpeDNet are elaborated as follows.

1) *SpaDNet:* Considering the spatial degradation representation of edge structure is more complex than the flattened area, we propose the SpaDNet with dual-branch parallel CNN architecture to represent spatial degradation. One branch is used to blur the whole observed image, and another branch focuses on edge structure processing. The dual-branch design can efficiently improve the nonlinear mapping ability to deal with more complex spatial degradation.
   The flowchart of the SpaDNet is shown in Fig. 2(a). The left branch employs an average pooling (AvgPool) to generate an LR-HSI with uniform blur. The right brand mainly focuses on edge structure processing. In the right brand, first, the structure information is extracted by a zero-mean Normalization (ZmNorm) [13] layer, which performs efficiently in passing high-pass zero-mean information. Then, the structure information is fed into the depth-wise convolutional layer (DW conv) to represent the edge structure degradation. After the DW conv, a Tanh activate function is employed for stronger nonlinear mapping ability. Finally, the spatially degraded image is obtained by adding the left and right output images. Since the AvgPool and ZmNorm can effectively keep the mean-value invariant between input and output images [13], the normalization characteristic of the spatial degradation is naturally kept.
2) *SpeDNet:* As (1), spectral degradation is a linear projection from HSI to MSI images along the spectral dimension. Hence, a fully connected layer can efficiently represent it, and the spectral degradation term of (1) can be rewritten by linear layer as:

$$\hat{\mathbf{Y}} = \mathbf{W}_S \hat{\mathbf{X}} \tag{5}$$

   where $\mathbf{W}_s \in \mathbb{R}^{3\times C}$ are the learnable weights of the linear layer.

Smoothness of $\mathbf{W}_s$ is crucial charastarastic for spectral degradation [28]. Previous works [31], [32] only employ a linear layer to represent the spectral degradation, which makes it difficult to constrain the smoothness of the spectral degradation. This article uses a 1-D convolutional layer (convolution size $= 3$) to learn the latent smoothness.

As shown in Fig. 2(c), $\mathbf{W}_s$ are fed into the Conv 1-D layer and ReLU activation function, and then the outputs are assigned as the new weights of a linear layer. The new weights of the linear layer will have smooth characteristics. The spectral degraded image can be calculated by:

$$\hat{\mathbf{Y}} = f_{\text{ReLU}}(f_{\text{Conv1D}}(\mathbf{W}_S))\hat{\mathbf{X}} \tag{6}$$

where $f_{\text{ReLU}}(\cdot)$ and $f_{\text{conv1D}}(\cdot)$ denote the ReLU activation and 1-D convolution, respectively.

B. FeafusFormer Architecture

In FeafusFormer, the multilevel features extracted from observed images are fused as the multilevel cross features. Consequently, these cross features are decoded as residual images superimposed on the upsampled LR-HSI to recover HR-HSI. This novel fusion framework fully uses the hierarchical feature at each level. It allows the network to be guided by the observers during the fusion, naturally ensuring high spatio-spectral similarity between the recovered and observed features. To fully model the global cross-similarity of multilevel features between the desired HR-HSI and the observed images, we propose the MCA block MCAB to encode the hierarchical features of observed HR-MSI and LR-HSI as cross-feature. A CNN-based decoder decodes the cross-feature of each level as residual images. The desired HR-HSI is obtained by superimposing these residual images onto the upsampled LR-HSI level by level.

The transformer-based multilevel feature encoding in MCAB and the CNN-based cross-feature decoding fully model the global and local cross-similarity between the desired HR-HSI and the observed images. This can effectively reduce spatio-spectral distortion and benefit HR-HSI recovery. Also, compared with the original observed images, the multilevel features have less spatial size, which can reduce the computational cost of the following cross-feature attention encoder.

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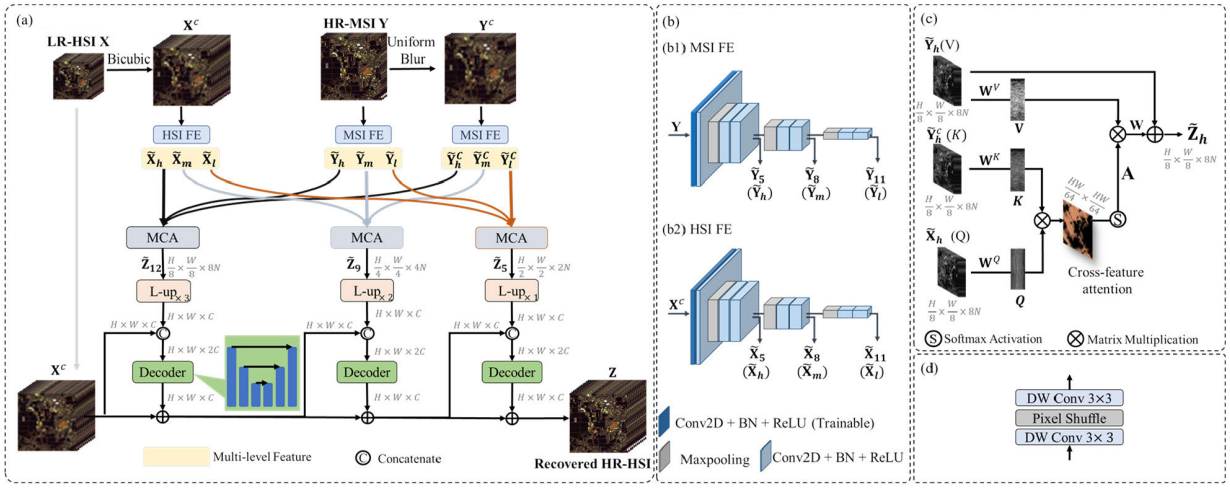


Fig. 3. Overall architecture of FeafusFormer. (a) By using MCAB, FeafusFormer realizes the multilevel fusion of observed images. (b) The architecture of feature extractor. (c) MCAB encodes the corresponding-level features as cross-feature at the stage of high-level cross feature encoding. (d) The architecture of the L-up model.

1) Flowchart: The flowchart of the FeafusFormer is shown in Fig. 3. As shown in Fig. 3(a), we first upsample the $\mathbf{X}$ as a coarse HR-HSI $\mathbf{X}^C \in \mathbb{R}^{W\times H\times C}$ in a bicubic manner, and uniformly blur the $\mathbf{Y}$ as a coarse HR-MSI $\mathbf{Y}^C \in \mathbb{R}^{W\times H\times 3}$. Then, we feed the $\mathbf{X}^C$, $\mathbf{Y}^C$, and $\mathbf{Y}$into the FeafusFormer. First, FeafusFormer exploits a well-trained feature extractor (FE) to extract the multilevel features of the observed images $[\tilde{\mathbf{X}}_l\ \tilde{\mathbf{X}}_m\ \tilde{\mathbf{X}}_h,\ \tilde{\mathbf{Y}}_l^c\ \tilde{\mathbf{Y}}_m^c\ \tilde{\mathbf{Y}}_h^c,\ \tilde{\mathbf{Y}}_l\ \tilde{\mathbf{Y}}_m\ \tilde{\mathbf{Y}}_h]$. Second, high-level features $[\tilde{\mathbf{X}}_h\ \tilde{\mathbf{Y}}_h^c\ \tilde{\mathbf{Y}}_h]$ are fed into the MCAB to generate high-level cross-features $\tilde{\mathbf{Z}}_h \in \mathbb{R}^{(W/8)\times(H/8)\times 8N}$, where N is the channel number of the FE. Third, $\tilde{\mathbf{Z}}_h$ undergoes three Learnable-upsample modules (L-up) for feature aggregation and upsample. As shown in Fig. 3(d), the L-up module is built up by a subpixel convolutional layer consisting of a DW conv with kernel size $= 3$, and a pixelshuffle layer. Subsequently, the upsampled high-level cross feature is concatenated with $\mathbf{X}^C$ and then is passed through the decoder to generate the high-level residual image $\mathbf{Z}^h \in \mathbb{R}^{W\times H\times C}$. The architecture of decoder is a U-Net with skip connections whose upsample and downsample layer is a stridded deconvolution 2×2 [34]. After superimposing $\mathbf{Z}^h$ onto $\mathbf{X}^C$, a coarse HR-HSI $\mathbf{Z}_h^C \in \mathbb{R}^{W\times H\times C}$, which contains high-level cross-feature, is obtained. Similarly, the middle-level residual image $\mathbf{Z}^m \in \mathbb{R}^{W\times H\times C}$ are generated and superimposed onto the $\mathbf{Z}_h^C$ to obtain a coarse HR-HSI $\mathbf{Z}_m^C \in \mathbb{R}^{W\times H\times C}$, which contains high-level and middle-level cross-features. Finally, the low-level residual image $\mathbf{Z}^l \in \mathbb{R}^{W\times H\times C}$ is generated, and superimposed onto the $\mathbf{Z}_m^C$ to obtain the desired HR-HSI $\mathbf{Z} \in \mathbb{R}^{W\times H\times C}$, i.e., $\mathbf{Z} = \mathbf{Z}^l + \mathbf{Z}_m^C$.

Next, we elaborate on the multilevel features extraction and MCAB.

2) Multilevel Features Extraction: This article employs the well-trained VGG-19 as a FE. VGG net [35] has demonstrated its superior feature extraction performance in style classification and texture recognition. The well-trained VGG-19 is trained from a large number of available MSI images by its designer for pattern recognition. The FE does not need to be retrained. So, using it for feature extraction in the fusion task is ingenious.

With the increasing depths of VGG-19 layers, it can extract different-level features. Fig. 3(b) displays the extraction process of the multilevel features. In the well-trained VGG19, the outputs at the front of each MaxPool layer are used as multilevel features. Since the well-trained VGG-19 only forward 3-channel images, the HSI and MSI (spectral channel larger than 3) cannot be fed directly into the VGG net. Therefore, as shown in Fig. 3, we modify the VGG-19 slightly as the MSI FE VGG^M and the HSI FE VGG^H. Specifically, we replace the first well-trained convolutional layer with a trainable one. The observed channel parameter of the trainable convolutional layer is set as the band number of the LR-HSI. Other parameter settings of the trainable layer are the same as the replaced layer. The multilevel features of the HR-MSI are represented as:

$$\tilde{\mathbf{Y}}_i = \mathrm{VGG}_{i-1}(\mathrm{Conv}(Y)) = \mathrm{VGG}_i^M(Y), \quad i = 5, 8, 11 \quad (7)$$

where $\mathrm{VGG}_i^M(\cdot)$ denotes the ith layer's output of the FE shown in Fig. 3(b1), $\tilde{\mathbf{Y}}_i$ is the ith-level feature of MSI. Multilevel features $\tilde{\mathbf{Y}} = [\tilde{\mathbf{Y}}_5, \tilde{\mathbf{Y}}_8, \tilde{\mathbf{Y}}_{11}]$, respectively, denote the low-level feature $\tilde{\mathbf{Y}}_l$, middle-level feature $\tilde{\mathbf{Y}}_m$, and the high-level feature $\tilde{\mathbf{Y}}_h$. The coarse HR-MSI multilevel features $\tilde{\mathbf{Y}}^c$ are extracted as (7).

To keep the same spatial dimension as the HR-MSI, LR-HSI $\mathbf{X}$ is upsampled as a coarse HR-HSI $\mathbf{X}^C$ in a bicubic manner. The LR-HSI multilevel features mainly consist of rich spectral features. The simple upsampling operation in the spatial domain has less effect on the spectral domain. As shown in Fig. 3(b2), the multilevel features $\tilde{\mathbf{X}}_i$ of the LR-HSI is obtained:

$$\tilde{\mathbf{X}}_i = \mathrm{VGG}_{i-1}(\mathrm{Conv}(X^c)) = \mathrm{VGG}_i^H(X^c), \quad i = 5, 8, 11. \quad (8)$$

3) Multilevel Cross-Feature Attention: Compared with traditional CNN, the transformer attention mechanism can exploit the global information more efficiently, which is beneficial for HR-HSI reconstruction. A transformer-based

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MCA block (MCAB) is proposed to efficiently model the global cross similarity. Using the MCAB, each level feature extracted from $\mathbf{X}$ and $\mathbf{Y}$ is encoded as a cross-feature. The cross-similarity of each level feature between the desired HR-HSI and observed images is explored.

Fig. 3(c) shows the MCAB used in the high-level cross-feature encoding. The high-level observed features $\tilde{\mathbf{X}}_h \in \mathbb{R}^{(W/8)\times(H/8)\times 8N}$, $\tilde{\mathbf{Y}}_h^C \in \mathbb{R}^{(W/8)\times(H/8)\times 8N}$, and $\tilde{\mathbf{Y}}_h \in \mathbb{R}^{(W/8)\times(H/8)\times 8N}$ are reshaped and linearly projected into query $\mathbf{Q} \in \mathbb{R}^{(WH/64)\times 8N}$, key $\mathbf{K} \in \mathbb{R}^{(WH/64)\times 8N}$, and value $\mathbf{V} \in \mathbb{R}^{(WH/64)\times 8N}$:

$$\mathbf{Q} = \mathbf{W}^{\mathbf{Q}}\mathbf{X}_h, \quad \mathbf{K} = \mathbf{W}^{\mathbf{K}}\tilde{\mathbf{Y}}_h^C, \quad \mathbf{V} = \mathbf{W}^{\mathbf{V}}\tilde{\mathbf{Y}}_h \tag{9}$$

where $\mathbf{W}^{\mathbf{Q}}$, $\mathbf{W}^{\mathbf{K}}$, and $\mathbf{W}^{\mathbf{V}} \in \mathbb{R}^{(W/8)\times(H/8)}$ are learnable parameters, the channel of features N is 64 determined in VGG-19.

The cross-feature attention map $\mathbf{A} \in \mathbb{R}^{(HW/64)\times(HW/64)}$ is calculated:

$$\mathbf{A} = \text{softmax}(\sigma \mathbf{K}\mathbf{Q}^T) \tag{10}$$

where $(\cdot)^T$ is transpose operation. $\sigma \in \mathbb{R}^1$ is a learnable parameter. Subsequently, the cross-feature attention is obtained by production of $\mathbf{V}$ and $\mathbf{A}$, i.e., $\mathbf{VA}$. Then, the cross-feature attention $\mathbf{VA}$ undergoes a linear projection and adds with $\mathbf{V}$. The high-level cross-feature $\tilde{\mathbf{Z}}_h$ is obtained:

$$\mathbf{MCAB}(\tilde{\mathbf{X}}_h, \tilde{\mathbf{Y}}_h^C, \tilde{\mathbf{Y}}_h) = \tilde{\mathbf{Z}}_h = \text{reshape}(\mathbf{VAW} + \mathbf{V}) \tag{11}$$

where $\mathbf{W} \in \mathbb{R}^{(W/8)\times(H/8)}$ are learnable parameters.

Similarly, middle-level cross-feature $\tilde{\mathbf{Z}}_m \in \mathbb{R}^{(W/4)\times(H/4)\times 4N}$ and low-level cross-feature $\tilde{\mathbf{Z}}_l \in \mathbb{R}^{(W/2)\times(H/2)\times 2N}$ is obtained by:

$$\tilde{\mathbf{Z}}_m = \mathbf{MCAB}(\tilde{\mathbf{X}}_m, \tilde{\mathbf{Y}}_m^C, \tilde{\mathbf{Y}}_m), \tilde{\mathbf{Z}}_l = \text{MCAB}(\tilde{\mathbf{X}}_l, \tilde{\mathbf{Y}}_l^C, \tilde{\mathbf{Y}}_l). \tag{12}$$

And the multilevel cross-feature $\tilde{\mathbf{Z}} = [\tilde{\mathbf{Z}}_h, \tilde{\mathbf{Z}}_m, \tilde{\mathbf{Z}}_l]$ is obtained.

IV. Training Details of uHNTC

The uHNTC comprises two degradation representation subnetworks: SpaDNet and SpeDNet, and a fusion subnetwork: FeafusFormer. We find that the initialization of degradation representation limits the recovery accuracy of FeafusFormer. Therefore, SpaDNet and SpeDnet are initialized at first. The degradation results of HR-HSI generated by FeafusFormer should be consistent with the observed images. According to this, the loss function of uHNTC is constructed. Through the unsupervised training of the uHNTC, the desired HR-HSI is obtained by FeafusFormer.

A. SpeDNet and SpaDNet Initialization

Before fusion, we first initialize the SpaDNet and SpeDNet. The LR-MSI generated by spatially degrading the HR-MSI $\mathbf{Y}$ should be consistent with the LR-MSI generated by spectrally degrading the LR-HSI $\mathbf{X}$. Therefore, the SpaDNet and SpeDNet can be initialized by following the absolute error (AE) loss function:

$$\mathcal{L}_0 = \|\text{SpaDNet}(Y, \theta_a) - \text{SpeDNet}(X, \theta_e)\|_1 \tag{13}$$

where $\|\cdot\|_1$ is the L1-norm.

Since the HSI spectral bands are highly correlated, and the spectral vectors often live in a low-dimensional subspace, it makes that the SpeDNet would only focus on several unrelated spectral bands, causing the SpeDNet to overfit. To solve this problem, we propose a subspace-based loss function. The SpeDNet can focus on all the spectral bands in the subspace where the components are orthogonal. Specifically, we employ singular value decomposition (SVD) to decompose the LR-HSI $\hat{\mathbf{X}}$ into subspace. And a spectral basis $\mathbf{\Psi} \in \mathbb{R}^{C\times k}$ and spatial coefficients $\mathbf{E} \in \mathbb{R}^{k\times WH}$ is obtained:

$$\hat{\mathbf{X}} = \Psi E. \tag{14}$$

As (1) and (15), the LR-MSI $\mathbf{Y}_{\mathbf{x}}$ is represented as:

$$\hat{\mathbf{Y}}_{\text{x}} = \text{SpeDNet}(\Psi, \theta_e)E. \tag{15}$$

The subspace-based loss function is represented as:

$$\mathcal{L}_s = \|\text{SpaDNet}(Y, \theta_a) - \text{SpeDNet}(\Psi, \theta_e)E\|_1. \tag{16}$$

Combining (14) and (17), the given loss function can efficiently initialize the SpaDNet and SpeDNet:

$$\mathcal{L}_1 = \mathcal{L}_0 + \frac{1}{2}\mathcal{L}_s. \tag{17}$$

B. Jointly Training

After initializing the SpaDNet and SpeDNet, the subsequent unsupervised joint training of uHNTC will help the FeafusFormer recover the desired HR-HSI better. Additionally, the SpaDNet and SpeDNet will be more accurate in representing spatial degradation and spectral degradation. There are three loss functions: the initialization loss $\mathcal{L}_1$, the data fidelity loss $\mathcal{L}_2$, and the perceptual loss $\mathcal{L}_3$.

1) Data Fidelity Loss: Data fidelity is that the recovered HR-HSI can be degraded as the observed LR-HSI and HR-MSI by the SpeDNet and SpaDNet, respectively. That is,

$$\mathcal{L}_2 = \left\|\text{SpaDNet}(Ff(X, Y, \theta_{Ff}), \theta_a) - X\right\|_1 + \left\|\text{SpeDNet}(Ff(X, Y, \theta_{Ff}), \theta_e) - Y\right\|_1. \tag{18}$$

2) Perceptual Loss: Perceptual loss has been widely used in style transfer to preserve content information [31]. In this article, we add a perceptual loss to improve the feature learning ability of the FeafusFormer. For the HSI-MSI fusion, we apply the perceptual loss between the generated HR-MSI and the observed HR-MSI. We employ the MSI FE $\text{VGG}^M(\cdot)$ as the FE in the perceptual loss. The perceptual loss $\mathcal{L}_3$ is defined as:

$$\mathcal{L}_3 = \sum_i \left\|\text{VGG}_i^M(\text{SpeDNet}(\text{Ff}(X, Y, \theta_{\text{Ff}}), \theta_e)) - \text{VGG}_i^M(Y)\right\|_1, \quad i = 2, 7, 14. \tag{19}$$

3) Final Loss Function: Ultimately, the overall loss function we used to train our networks is given by:

$$\mathcal{L}_{\text{final}} = \mathcal{L}_1 + \mathcal{L}_2 + \lambda\mathcal{L}_3 \tag{20}$$

where λ is a balance parameter, which is set as 0.1. By backpropagation in the network training of uHNTC, the θ_{Ff}, θ_a, and θ_e is automatically updated. And the desired HR-HSI is recovered by using FeafusFormer as (2).

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Algorithm 1 Proposed uHNTC Algorithm

Input: HR-MSI $\mathbf{Y}$ and LR-HSI $\mathbf{X}$

Procedure:

1. Initialize the SpaDNet and SpeDNet:
   (a) Optimize the weights θ_a in SpaDNet and θ_e in SpeDNet using (17)
2. Blind HSI-MSI fusion
   (a) Solve the weights θ_{Ff} in FeafusFormer, θ_a in SpaDNet, and θ_e in SpeDNet using (20)
   (b) Obtain the desired HR-HSI using (2)

End Procedure

Output: the desired HR-HSI $\mathbf{Z}$

V. Experiments

A. Datasets

In this study, to verify the performance of the proposed uHNTC, we adopt three public datasets, i.e., PaviaU dataset [49], Chikusei dataset [50], Xiongan dataset,[1] and real-world WorldView-2[2] images. The public datasets are used for simulated experiments, and the real-world images are used for real-world experiments. These datasets are detailed as follows.

1) PaviaU: In the Pavia University dataset, 115 bands cover the spectral range from 430 to 860 nm with an interval of 10 nm. It contains 610 × 340 pixels from nine classes of materials. Following [11], after removing the water vapor absorption bands, the remaining 93 bands are used in this study.

2) Chikusei: There are 128 bands, and the scene consists of 2517 × 2235 pixels. The HSI covers the wavelength range of 363–1018 nm.

3) Xiongan: There are 250 bands, and the scene consists of 3750 × 1580 pixels. The HSI covers the wavelength range of 400–1000 nm.

4) World View-2: The images acquired by the WorldView-2 satellite only contain pairs of the HR Panchromatic images (HR-PAN) with a size of 1692 × 1692 and the LR-MSI with a size of 423 × 423 × 4.

B. Experimental Settings

In simulated experiments, the HSI from the public dataset is set as the ground truth. Following [52], the IKONOS-like reflectance spectral response filter is used to simulate the spectral degradation, and the HR-MSI is generated as (1). The Gaussian-based downsample is employed to simulate the spatial degradation, and the LR-HSI is generated as (1). We set the Gaussian kernel as 33, and the standard deviation as 22.

To evaluate the performance of the proposed method, uHNTC is compared with ten state-of-the-art HSI fusion methods. The compared methods include four optimization-based methods: subspace regularized method (HySure [32]), low tensor multirank-based method (LTMR [34]), CNN denoiser-based method (CNN-Fus [40]), multischeme-based method (HyMS [19]), and six deep-learning based method: model-guided deep convolutional network method (MoG-DCN [16]), deep blind superresolution method (DBSR [20]), unsupervised adaptively learning-based method (UDALN [21]), fusion transformer (FusFormer [12]), pyramid shuffle-and-reshuffle transformer (PSRT [14]), and multiscale spatio-spectral transformer (MSST [30]). The FusFormer, PSRT, and the MSST are the state-of-the-art transformer models. The HySure, DBSR, MoG-DCN, and UDALN are blind methods. The LTMR and CNN-Fus are nonblind methods. The HyMS is a semiblind method.

TABLE I
Size of the Testing Images

Dataset	Ground Truth	LR-HSI	HR-MSI	r
Pavia	320×320×93	10×10×93	320×320×4	32
Chikusei	512×512×128	16×16×128	512×512×4	32
Xiongan	512×512×250	16×16×250	512×512×4	32
World View-2	——	400×400×4	1600×1600×1	4

On the Pavia University dataset, the bottom-left region with a spatial resolution of 320 × 320 is served as a testing image. On the Chikusei and Xiongan datasets, we pick the top-left region with a spatial size of 512 × 512 as testing images.

For the supervised methods (MoGDCN, FusFormer, PSRT, and MSST), the remaining regions are clipped as 128 × 128 overlapped patches to form training datasets. In the real-world experiment, since the ground truth of the HR-MSI is unavailable, we follow the Wald's protocol [51] to form the training samples. A total of 2048 pairs of 200 × 200 × 1 HR-PAN patches and 50 × 50 × 4 LR-MSI patches are generated, and the origin LR-MSIs are set as the ground truth. Table I reports testing images' size for clear representation, i.e., the ground truth, LR-HSI, HR-MSI, and the scale factor r.

Five quality metrics, i.e., peak signal-to-noise ratio (PSNR) [19], spectral angle mapper (SAM) [19], relative dimensionless global error in synthesis (ERGAS) [52], structural similarity index (SSIM) [52], and root mean square error (RMSE) [52] are used to measure the performance of all methods quantitatively. Unlike the PSNR and SSIM (the larger, the better), the ERGAS, SAM, and RMSE are inversely proportional to the image quality (the smaller, the better).

During training, we employ AdamW optimizer with an initial learning rate of 8e-3 and decay the learning rate with gamma 0.99 every 100 epochs. The max iteral epoch is set as 5000. We evaluate all the testing methods on a computer with an Nvidia RTX 3090 GPU and an Intel Xeon Platinum 8168 CPU.

C. Ablation Study

1) Subspace-Based Loss Function: We conduct an ablation study on the effectiveness of the subspace-based loss function in (16). The quantitative results with different loss function schemes on the Chikusei dataset are listed in Table II. The optimal results are bold. In Table II, S1 means the SpeDNet and SpaDNet are not initialized. S2 denotes the SpeDNet and SpaDNet are initialized using the AE loss function.

[1] http://www.hrs-cas.com/a/share/shujuchanpin/2019/0501/1049.html
[2] https://liangjiandeng.github.io/PanCollection.html

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TABLE II
QUANTITATIVE RESULTS PRODUCED BY DIFFERENT INITIALIZATION SCHEMES

Scheme	PSNR	SAM	RMSE
S1	19.33	24.31	30.20
S2	42.39	3.99	2.01
S3	39.27	6.60	3.69
S2+S3	**46.00**	**2.84**	**1.57**

TABLE III
ABLATION STUDY ON PERCEPTUAL LOSS

Perceptual Loss	PSNR	SAM	RMSE
×	47.26	2.54	1.24
√	**48.05**	**2.42**	**1.13**

TABLE IV
ABLATION STUDY ON MULTILEVEL FEATURE

Low	Middle	High	PSNR	SAM	RMSE
√	×	×	44.91	3.01	1.63
√	×	√	45.62	2.89	1.48
√	√	×	46.20	2.74	1.37
√	√	√	**48.05**	**2.42**	**1.13**

S3 means SpeDNet and SpaDNet are initialized using the subspace-based loss function. S2 + S3 means the SpeDNet and SpaDNet are initialized using the AE loss function and the subspace-based loss function. Obviously, the S2 + S3 scheme can effectively improve recovery accuracy, which demonstrates the effectiveness of the proposed subspace-based loss function in benefit training our neural network.

2) Perceptual Loss Function: To demonstrate the effectiveness of perceptual loss function, we remove this loss function, and the experimental results on the Chikusei dataset are shown in Table III. It can be seen that the perceptual loss function can help the FeafusFormer recover a better HR-HSI.

3) Multilevel Feature Fusion Framework: In this study, the effectiveness of multilevel feature fusion scheme is investigated by removing partly observes' hierarchical features during fusion. Table IV lists the quantitative results on the Chikusei dataset, and the optimal results verify the effectiveness of the multilevel feature fusion framework.

4) Multilevel Cross-Feature Attention: MCA is the core fusion unit of FeafusFormer. To evaluate the effectiveness of MCA, we investigate the performance of FeafusFormer by removing MCA. For the FeafusFormer without MCA, the hierarchical features are concatenated and fed into the decoder. The quantitative results on the Chikusei dataset are shown in Table V. It can be seen that MCA improves the FeafusFormer significantly.

5) SpaDNet and SpeDNet: The SpaDNet and SpeDNet play an important role in blind HSI-MSI fusion. In this section, we replace the SpaDNet and SpeDNet in uHNTC with different degradation methods (HySure, UDALN, and the DBSR) to verify the effectiveness of proposed degradation representation networks. Table VI shows the experimental results on the Chikusei dataset. The optimal quantitative results demonstrate the superiority of the proposed degradation networks.

TABLE V
ABLATION STUDY ON MULTILEVEL CROSS-FEATURE ATTENTION

MCA	PSNR	SAM	RMSE
×	44.12	3.11	1.65
√	**48.05**	**2.42**	**1.13**

TABLE VI
ABLATION STUDY ON DEGRADATION NETWORKS

Method	PSNR	SAM	RMSE
with HySure	44.79	3.12	1.98
with UDALN	45.32	3.06	1.74
with DBSR	45.30	3.06	1.74
Ours	**46.00**	**2.84**	**1.57**

D. Performance on Public Datasets

1) Blind Degeneration: The spatial and spectral degradation models are assumed unknown in this experiment. In the test images, following [20], we employ a motion blur kernel as the spatial degradation, and the IKONOS-like SRF with white noise reaches 50 dB as the spectral degradation. We employ the Gaussian-based downsample for the supervised methods as the spatial degradation and the IKONOS-like SRF as the spectral degradation to form training samples. For the nonblind methods (i.e., LTMR and CNN-Fus), following [16], we give them a predefined spatial degeneration (a Gaussian blur kernel of size 9 × 9 and deviation 1.5) and a predefined spectral degeneration (adding Gaussian white noise into the SRF until the SNR reaches 40 dB).

The experimental results on the PaviaU and Chikusei datasets are reported in Table VII. We highlight the optimal and suboptimal results in bold and underlined, respectively. Ours performs the highest recovery accuracy in terms of the five metrics. In specific, our PSNR reaches 46.00 dB, and the SAM is only 2.84, which greatly pushed forward the limits at the 32× super-resolution fusion. In the same way, ours still maintains the highest performance on the PaviaU dataset. Nonblind methods perform unsatisfactory since they receive imprecise degradation models. Supervised methods MoGDCN, Fusformer, MSST, and PSRT cannot efficiently achieve HSI-MSI fusion, which is caused by the degradation differences between the training and testing images. Due to the differences in image statistics between training and test images, there is a so-called domain shift [53] between the learned image domain and the target one, generally leading to a significant performance drop. To demonstrate this view, in Fig. 4, we depict the PSNR which varies in the learned and target image domains. In Fig. 4, the learned image domain indicates that the spatial and spectral degradation in the test image is consistent with the training images. And the target image domain indicates that the spatial and spectral degradation in the test image is inconsistent with the

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TABLE VII
QUANTITATIVE PERFORMANCE OF 11 METHODS ON PAVIAU AND CHIKUSEI DATASETS WITH BLIND DEGRADATIONS

	PaviaU Dataset					Chikusei Dataset				
	PSNR	SAM	ERGAS	SSIM	RMSE	PSNR	SAM	ERGAS	SSIM	RMSE
HySure	37.00	3.71	**0.25**	0.988	3.54	42.87	3.55	0.48	0.985	1.95
LTMR	24.05	13.26	1.27	0.817	16.26	31.22	15.62	2.00	0.782	7.65
CNN-Fus	23.82	13.68	1.28	0.817	18.09	32.70	12.12	1.83	0.818	6.48
HyMS	35.33	4.70	0.36	0.975	5.23	45.42	3.06	0.44	0.989	1.61
MoG-DCN	28.48	4.19	0.78	0.960	9.75	36.06	6.50	1.20	0.950	4.34
Fusformer	30.63	4.51	0.64	0.973	7.66	35.06	6.71	1.23	0.944	4.91
MSST	27.94	5.99	0.92	0.941	10.51	35.49	7.17	2.93	0.935	4.99
PSRT	26.34	5.14	1.02	0.940	12.38	33.86	7.89	1.87	0.902	6.07
DBSR	32.92	5.93	0.54	0.970	5.54	40.65	4.28	0.61	0.978	2.42
UDALN	36.47	3.51	0.39	0.980	5.59	41.78	3.13	0.53	0.983	2.28
Ours	**38.04**	**3.30**	**0.25**	**0.989**	**3.38**	**46.00**	**2.84**	**0.39**	**0.993**	**1.57**

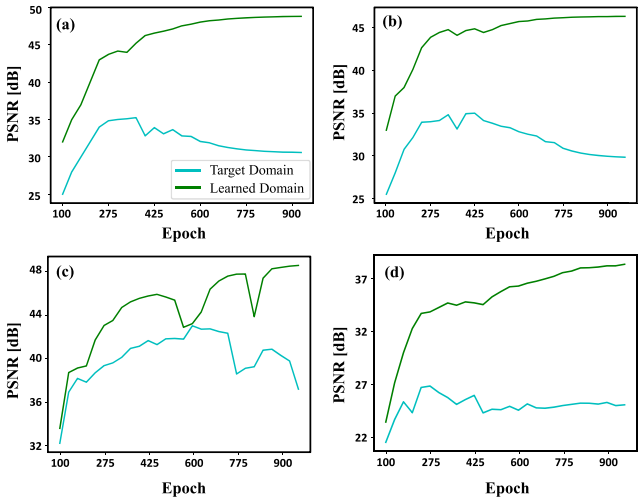


Fig. 4. PSNR varies in terms of epochs in the learned image domain and the target image domain on the Xiongan dataset. (a) MoGDCN. (b) MSST. (c) Fusformer. (d) PSRT.

TABLE VIII
QUANTITATIVE PERFORMANCE OF 11 METHODS ON XIONGAN DATASET WITH BLIND DEGRADATIONS

	Xiongan Dataset				
	PSNR	SAM	ERGAS	SSIM	RMSE
HySure	41.78	2.12	0.18	0.985	2.29
LTMR	38.96	3.81	0.27	0.954	4.24
CNN-Fus	39.44	2.66	0.20	0.971	3.09
HyMS	38.74	1.60	0.20	0.993	2.75
MoG-DCN	30.91	2.01	0.51	0.973	8.46
Fusformer	38.31	1.42	0.27	0.984	2.85
MSST	30.22	2.74	0.55	0.970	8.98
PSRT	25.58	4.03	1.02	0.831	17.54
DBSR	39.85	1.94	0.37	0.990	2.84
UDALN	38.39	2.48	0.44	0.983	3.17
Ours	**47.96**	**1.25**	**0.17**	**0.995**	**1.45**

training images. From Fig. 4, the deep networks achieve better performance in the learned image domain with the epochs increasing. Conversely, they become worse in the target image domain. This finding is consistent with the conclusion that the degradation information is stationary once the network is trained successfully, reducing their generalization [19].

On the Xiongan dataset, Table VIII reports the quantitative performance of different methods. Our method still remains the best recovery. The performance of nonblind model-based methods LTMR and CNN-Fus is discontent. The imprecise degradation models limit the nonblind model-based methods, and the supervised methods suffer from the domain shift.

To visualize the performance for spectral reconstruction, Fig. 5 shows the AE maps between the recovered and ground truth bands. The AE maps produced by different methods at the 10th and 60th bands on the PaviaU dataset are shown in the first two rows, respectively. The corresponding error maps on the Chikusei dataset at the 42nd and 69th bands are shown in the 3rd and 4th rows, respectively, and the results on the Xionan dataset at 35th and 92nd bands are shown in the last two rows. Obviously, ours produces minor errors, while the compared methods almost produce a significant error, which indicates that ours also performs the best detail recovery.

2) Known Degeneration: The proposed uHNTC can achieve blind fusion tasks. Also, the fusion network FeafusFormer performs the best in coping with known degradation. We evaluate the performance of our proposed method in dealing with known degradation. The true spatial degradation and spectral degradation models in this subsection are known for each testing method, and the degradation models mentioned in Section IV-B are employed as the true ones.

Table IX lists the quantitative performance of eleven methods on the PaviaU and the Chikusei datasets. Our method performs better than compared methods according to all quality metrics, which indicates that HR-HSIs recovered by our method have the best spatial and spectral qualities. For example, on the Chikusei dataset, our PSNR values are higher than the second-place MoGDCN by about 1 dB. Moreover, our PSNR and SSIM values achieved 48.05 and 0.995, respectively. It has a significant improvement over the compared

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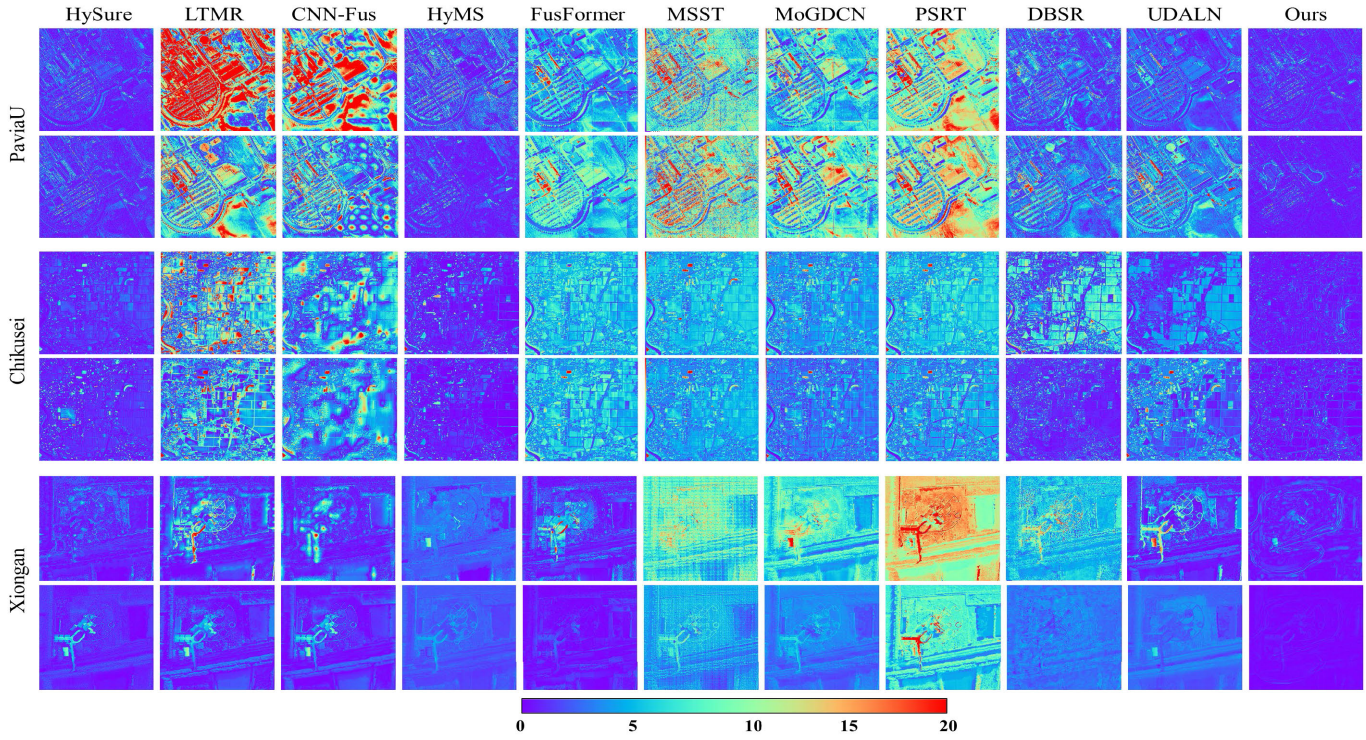


Fig. 5. Absolute error maps between blind fusion results and ground truth on the PaviaU dataset (the first two rows), Chikusei dataset (third and fourth rows), and the Xiongan dataset (the last two rows).

TABLE IX
QUANTITATIVE PERFORMANCE OF 11 METHODS ON PAVIAU AND CHIKUSEI DATASETS WITH KNOWN DEGRADATIONS

	PaviaU Dataset					Chikusei Dataset				
	PSNR	SAM	ERGAS	SSIM	RMSE	PSNR	SAM	ERGAS	SSIM	RMSE
HySure	39.99	2.71	0.25	0.993	2.96	45.03	3.55	0.44	0.990	1.52
LTMR	33.89	8.36	0.54	0.928	8.02	38.23	9.63	1.05	0.930	4.38
CNN-Fus	37.99	5.06	0.33	0.978	4.85	42.33	5.51	0.74	0.976	2.46
HyMS	39.28	3.24	0.27	0.987	3.62	46.00	2.79	0.42	0.991	1.33
MoG-DCN	39.07	2.41	0.23	0.992	2.86	47.58	2.50	0.40	0.993	1.25
Fusformer	41.06	2.56	0.23	0.993	2.39	47.40	2.45	0.37	0.994	1.28
MSST	36.62	4.58	0.42	0.982	4.94	46.43	2.97	0.42	0.992	1.30
PSRT	38.71	2.61	0.24	0.991	2.99	46.10	2.57	0.39	0.993	1.32
DBSR	35.60	4.35	0.32	0.981	4.58	43.50	3.47	0.44	0.989	1.77
UDALN	40.14	2.55	0.19	**0.995**	2.51	45.94	2.87	0.40	0.991	1.36
Ours	**41.45**	**2.30**	**0.17**	**0.995**	**2.30**	**48.05**	**2.42**	**0.34**	**0.995**	**1.13**

methods. On the PaviaU dataset, our PSNR value is higher at 1.31 dB than the second-place UDALN. Besides, compared with the results of blind degradation, the performance of the nonblind methods (LTMR and CNN-Fus) and supervised methods (MoGDCN, Fusformer, MSST, and PSRT) are significantly improved, demonstrating that their performance is limited in coping with blind degradation. Therefore, it is necessary to investigate the blind fusion method.

Table X summarizes the quantitative results of all the methods on the Xiongan dataset. Ours produce the best quantitative result compared to competing methods. It indicates that our method keeps superior performance in coping with known degradation. Our PSNR value has achieved 51.22 dB, higher than the second-place MoGDCN and HyMS by 2.3 dB. Our SAM value is only 0.79, which is lower than 0.09 for the second-place MoGDCN, demonstrating that ours has the best performance of the spectral recovery.

To better visualize the recovery results, the AE maps and SAM error maps are shown in Fig. 6. In the SAM error map, each pixel value represents the SAM value between the recovered spectrum and the ground truth spectrum. An edge region is selected and zoomed in the corresponding box for precise observation. The first three rows show the AE maps at 39th, 60th, and 80th bands, respectively. From the first three rows, we can see that our method produces fewer errors, which indicates that our method performs the best. The last

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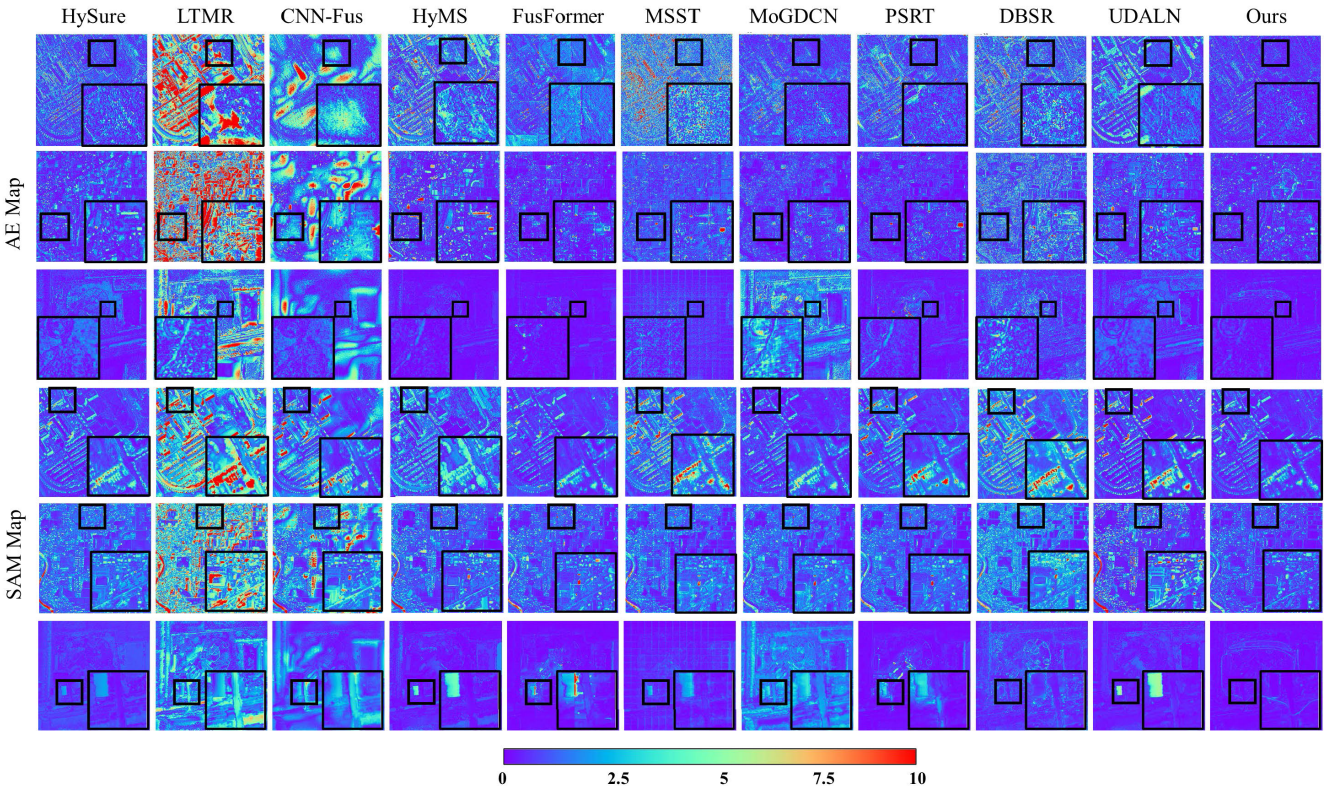


Fig. 6. Absolute error maps (the first three rows) and SAM map (the last three rows) produced by different methods on public datasets. The first and fourth rows show the results on the PaviaU dataset; the second and fifth rows show the results on the Chikusei dataset; and the third and last rows show the results on the Xiongan dataset.

TABLE X
QUANTITATIVE PERFORMANCE OF 11 METHODS ON XIONGAN DATASET WITH KNOWN DEGRADATIONS

	Xiongan Dataset				
	PSNR	SAM	ERGAS	SSIM	RMSE
HySure	44.89	1.79	0.12	0.993	1.69
LTMR	39.44	2.57	0.20	0.971	3.08
CNN-Fus	41.65	2.45	0.18	0.980	2.83
HyMS	48.82	1.20	0.08	0.995	1.36
MoG-DCN	48.82	0.87	0.07	0.997	1.15
Fusformer	48.54	0.88	0.08	0.996	1.35
MSST	47.42	1.31	0.08	0.995	1.38
PSRT	39.35	0.96	0.13	0.965	4.47
DBSR	43.69	1.98	0.15	0.987	2.05
UDALN	46.30	1.25	0.09	0.993	1.50
Ours	**51.22**	**0.79**	**0.05**	**0.998**	**0.90**

three rows display the SAM error maps. Our SAM map shows fewer errors from the global and local views. On the PaviaU dataset, ours has the best performance, and the HySure is the second. The LTMR and DBSR produce large SAM errors, which indicates that the recovery spectrum has a significant error. And on the Chikusei dataset, especially in the lower left corner, only ours can recover the spectrum of the river, while others produce tremendous SAM errors. On the Xiongan dataset, ours and DBSR successfully recovered the spectra of buildings. And ours produce less error in the edge region. Our method achieves the best recovery in spectral and spatial domains compared to ten competing methods.

E. Computational Cost

To analyze the computational cost of all the compared methods, the number of the floating-point operations per second (FLOPs), and the trainable params on the Xiongan dataset are listed in Table XI. It can be seen that, as a Transformer-based method, ours yields the most significant improvement with the appropriate computational cost.

F. Performance on WorldView-2 Images

In the real-world experiment, the PAN-MSI pairs captured by WorldView-2 are used to test the practicability of our method. Since degradation models are required for nonblind methods (LTMR and CNN-Fus), we employ the spectral and spatial degradations solved in HySure [34] as their predefined degradation models.

The spectral band images recovered by different methods are shown in Fig. 7. The meaningful regions are magnified in red boxes for a clear comparison. Notably, the results of our method and those of DBSR, UDALN, and FusFormer are markedly clearer, whereas the other competing methods lack fine-grained reconstruction on agricultural land and buildings. Moreover, the brightness of ours is closer to that of the

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TABLE XI
COMPUTATIONAL ANALYSIS OF THE DEEP-LEARNING BASED METHODS ON XIONGAN DATASET

	UDALN	DBSR	MoGDCN	FusFormer	PSRT	MSST	Ours
FLOPs	57.16 G	3.26 T	4.95 T	37.01 G	76.67 G	2.43 T	422.85 G
Param	8.89 K	12.71 M	6.86 M	141.7 K	283.52 K	65.20 M	6.74 M

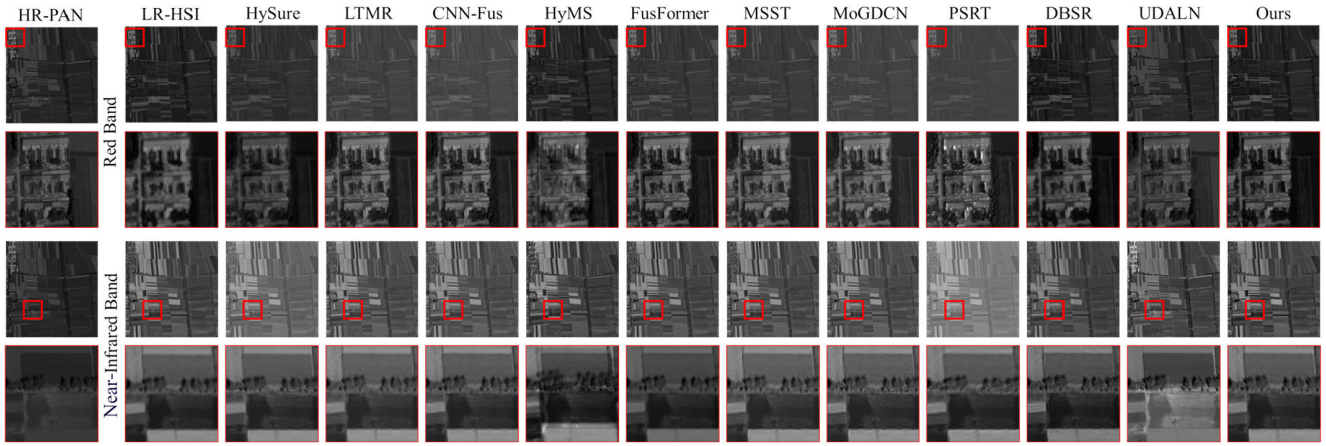


Fig. 7. Experimental results on real-world images. Some details are depicted in the red rectangles. The first two rows show the recovered results on the red band; and the last two rows show the recovered results on the near-infrared band.

LR-HSI, indicating ours recovers better spectral accuracy. In comparison, our method performs well in terms of the reconstruction of detail and intensity, demonstrating that the proposed method outperforms the state-of-the-art methods in practice.

VI. CONCLUSION

For blind HSI-MSI fusion, an uHNTC is proposed. It comprises three subnetworks, i.e., SpaDNet, SpeDNet, and FeafusFormer.

Following the characteristics of degradation information, two CNN-based subnetworks, SpaDNet and SpeDNet, are designed to represent the spatial and spectral degradations, respectively. In the SpaDNet, a novel dual-branch parallel CNN architecture is designed, which enables the SpaDNet to pay well attention to the degradation of structural regions. Subsequently, SpeDNet employs a 1-D convolutional layer to adjust the weights in a linear layer, which can efficiently exploit the latent smooth characteristics of the spectral degradation. Also, to further exploit the sparsity and smoothness of the degradation information, we investigate a subspace-based loss function, significantly improving the representation accuracy of degradation networks.

By fully exploring the cross-interactions in the desired HR-HSI, a feature fusion transformer (FeafusFomer) is designed to improve recovery accuracy. In the FeafusFormer, the multilevel feature fusion framework and the proposed MCAB give FeafusFormer a stronger ability to exploit the hierarchical feature's local and global information. Also, the MCAB efficiently explores the cross-similarity between the desired HR-HSI and observed images, thereby ensuring the cross-interactions in the recovery HR-HSI, efficiently reducing the spatio-spectral distortion. Through a joint training scheme, uHNTC solves the blind HSI-MSI fusion problem excellently, recovering the HR-HSI with very high accuracy. Experiments on three public HSI datasets and real-world WorldView-2 satellite images demonstrate the superior performance of uHNTC in coping with blind and nonblind HSI-MSI fusion.

In addition, the uHNTC still has ample room for improvement. To begin with, SpaDNet employs a fixed degradation process for each pixel in an image due to the translation invariance of the convolutional layer. Typically, the inherent property of imaging sensors always leads to spatial variations in the observed image, indicating that each pixel possesses its own degradation matrix. Consequently, exploring an adaptive degradation network could enhance the accuracy of blind fusion. Furthermore, the U-shaped Decoder in the FeafusFormer might not efficiently decode the fused hierarchical features. Integrating the feature decoding into the MCA could enhance the performance of the FeafusFormer.

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Xuheng Cao (Student Member, IEEE) received the M.E. degree in light industry technology and engineering from the Beijing Institute of Graphic Communication, Beijing, China, in 2023. He is currently pursuing the Ph.D. degree in optical engineering with the Institute of Precision Optical Engineering, Tongji University, Shanghai, China.

He is also a member of the Remote Sensing Calibration Laboratory, National Institute of Metrology, Beijing, China. His research focuses on computational imaging and infrared remote sensor calibration.



Yusheng Lian received the Ph.D. degree from the National Laboratory of Color Science and Engineering, Beijing Institute of Technology, Beijing, China, in 2014.

He is currently a Professor with the School of Printing and Packaging Engineering, Beijing Institute of Graphic Communication, Beijing, China. His research interests include the multispectral imaging and image processing technology.



Kaixuan Wang received the B.Eng. degree from the Faculty of Printing and Packaging Engineering, Beijing Institute of Graphic Communication, Beijing, China, in 2022, where she is currently pursuing the master's degree in image science and engineering.

Her research interests include image halftoning and image processing.



Chao Ma received the B.Eng. degree from the School of Printing and Packaging Engineering, Beijing Institute of Graphic Communication, Beijing, China, in 2022, where he is currently pursuing the master's degree in image science and engineering.

His research interests include hyperspectral compressed sensing imaging.



Xianqing Xu received the B.Eng. degree from the School of Printing and Packaging Engineering, Beijing Institute of Graphic Communication, Beijing, China, in 2023, where he is currently pursuing the master's degree in image science and engineering.

His research interests include deep learning and image fusion.

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